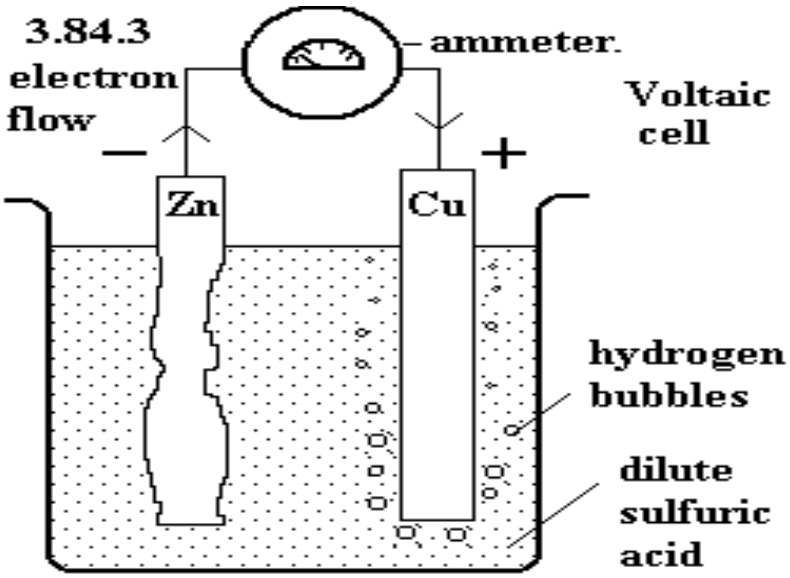
Wet cells, Dry cells and Batteries

The **electrochemical cell**.

A simple device that converts stored chemical energy into electrical energy. A **battery** is one or more electrochemical cells together. Alessandro Volta invented the first battery (or "voltaic pile") in 1800.

Wet Cell

A wet cell has a couple of parts



The electrodes. These are plates of _____ or _____. One plate breaks down and gives away electrons. The other plate collects these _____.

The plate that gives away electrons us called the **Anode**.

The plate that collects electrons is called the **Cathode**

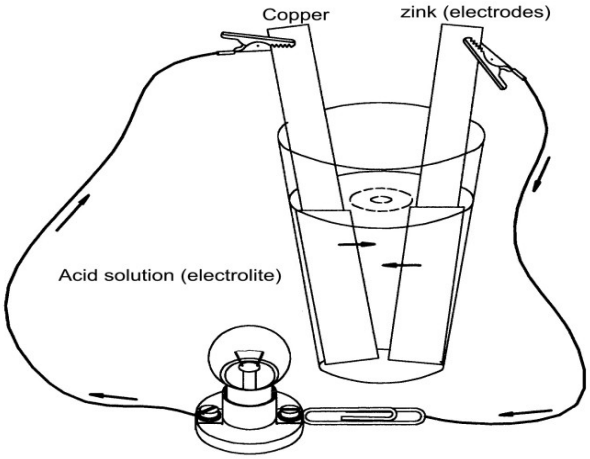
An Electrolyte. This is a _____that _____electricity. This movement of electrons gives a current.

A wet cell is not very portable as the electrolyte can _____. Also as the anode gives away its electrons it breaks down. Once the anode has completely broken down the wet cell will stop working.

An example of a wet cell used today is :_____

_____

Label the electrodes and electrolyte in the diagram below.



What were the problems with a wet cell?

Dry Cell

A dry cell has the same basic parts as a wet cell, but has characteristics that make it more useful. The electrolyte is a _____, not a liquid. As a result it does not splash around and spill if the cell is moved. The zinc electrode has been made into part of the outer casing of the battery. The other electrode is often made of graphite.

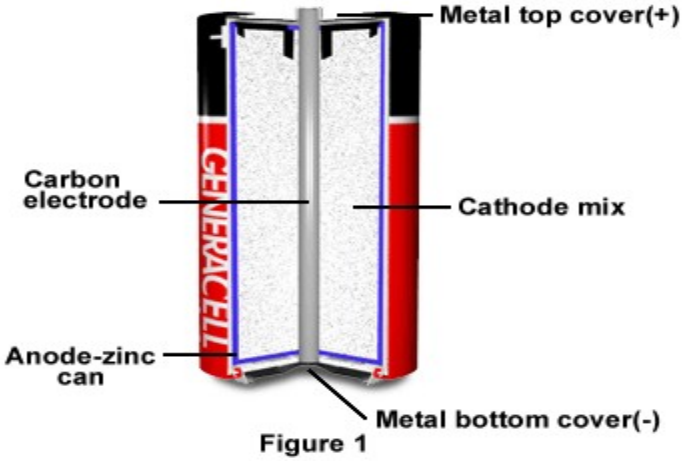


Figure 1

Why is the dry cell so useful?

Photovoltaic Cells

Another word for photovoltaic cell is _____ cell. These convert solar energy directly into _____ energy. The solar energy knocks _____ off of the silicon crystals inside. As the electrons move away they form an electric current.

The current can be used straight away or it can be used to charge a battery for later use.

Primary and Secondary Batteries

Primary batteries (disposable batteries). These are designed to be used only once and then thrown away. This is a problem as the batteries must be replaced regularly. Also the wastes in these batteries are toxic, so safe disposal is not always easy.

Secondary batteries (rechargeable batteries or storage cells). These are designed to be recharged and used many times.

Questions:

1. A mobile phone uses a _____ cell _____ battery, because it has a paste electrolyte and can be recharged.

2. Bob has a torch that he uses when he goes camping. After a few nights use the globe gets quite dim. Bob replaces the torch batteries with new ones and safely disposes of the old ones. The batteries in Bob's torch where _____ cells and _____ batteries.

3. A car battery has a liquid electrolyte so it is a type of _____ cell. It can be recharged, so it is a _____ battery.

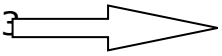
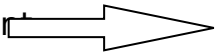
4. What type of cell/battery would be best to use in an MP3 player? Give a reason for your answer.

5. Why must you be careful when carrying a car battery?

6. Why must batteries be disposed of carefully?

7. Energy conversion= Where energy changes from one _____ of energy to another.

8. Draw the energy conversion flow diagram for the following.

Battery in Mp3  player current  music produced

9. Complete this table

Word	Meaning
Battery	
Wet Cell	
Dry Cell	
Electrode	
Anode	
Cathode	
Electrolyte	